

Kafil against najm-theme

A source-verified adoption review of Kafil's appearance, branding, preset, asset, MCP, and server-rendering contracts against the published najm-theme 0.1.1 package.

KAFIL SOURCE	NAJM SOURCE	REVIEW MODE
HEAD e1298c5 plus a pre-existing dirty worktree	HEAD 85634d2; najm-theme 0.1.1	No runtime or repository changes

Corrected outcome

9	3	13
ALIGNED	CONFIGURABLE	BREAKING
Same behavior or PostgreSQL contract.	Parity through package configuration.	Migration, client, wire, route, or automation change.

Headline

Adoption is not a redesign. Kafil independently reached the same revision protocol, reset semantics, fallback behavior, candidate lifecycle, standard slot graph, and module-scope RSC architecture. The work is concentrated in persistence and wire-shape migration.

Freeze blockers

Do not adopt yet. First review School, then freeze: appearance merge granularity; preset collision policy; null branding values; built-in preset deletion; actor-attribution integrity. The Kafil worktree must also be made reproducible before implementation begins.

Twenty-five verified comparison points

The counts below are generated from the entries actually present: 9 aligned, 3 configurable, and 13 breaking.

Configuration and transport

1. Appearance and branding route prefix

CONFIGURABLE

KAFIL

/api/appearance and /api/branding under Server.base('/api').

NAJM-THEME 0.1.1

basePath: " produces /appearance and /branding under the same server base.

Impact. External appearance and branding routes remain unchanged.

2. Preset routes

BREAKING

KAFIL

/api/theme-presets and /api/theme-presets/:id/apply.

NAJM-THEME 0.1.1

/api/presets and /api/presets/:id/apply when basePath is empty.

Impact. Update the preset API, query keys, hooks, tests, and any saved links.

3. MCP names and surface

BREAKING

KAFIL

Controller-derived appearance and preset tool groups with Kafil confirmation copy.

NAJM-THEME 0.1.1

Five theme_* tools; no appearance-save tool and no binary upload tool.

Impact. Saved prompts and automation must move to the package tool names. Upload stays REST-only.

4. Public appearance and branding reads

ALIGNED

KAFIL

GET reads are public so login and first-login chrome can render.

NAJM-THEME 0.1.1

publicRead: true applies the same policy explicitly.

Impact. No authorization-policy change for public bootstrap reads.

5. Feature set

CONFIGURABLE

KAFIL

Appearance, branding, presets, uploads, and MCP are all enabled.

NAJM-THEME 0.1.1

The same five capabilities are explicit feature flags.

Impact. Enable all five; partial-feature behavior remains available for tests and future consumers.

Persistence and identity

6. Storage provider

BREAKING

KAFIL

Direct filesystem writes in Kafil's branding directory.

NAJM-THEME 0.1.1

najm-storage >= 2.2.0, resolved by stable token and scoped namespace.

Impact. Add and configure storage before theme(). Decide the production provider explicitly.

7. Table layout

BREAKING

KAFIL

Appearance and branding occupy seven columns on platform_settings; presets use theme_presets.

NAJM-THEME 0.1.1

najm_theme_appearance, najm_theme_branding, and najm_theme_presets.

Impact. Use an additive backfill. Do not drop legacy columns in the cutover release.

8. Branding persistence shape

BREAKING

KAFIL

Four nullable *_path columns.

NAJM-THEME 0.1.1

A slot_config JSON map keyed by registered slot name.

Impact. Convert paths to asset records. This enables custom slots without schema changes.

9. Actor foreign keys

BREAKING

KAFIL

updated_by_user_id and created_by_user_id reference users with ON DELETE SET NULL.

NAJM-THEME 0.1.1

updated_by_actor_id and created_by_actor_id are unconstrained text.

Impact. Freeze whether historical attribution may outlive a deleted user or whether Kafil adds an app-owned integrity layer.

10. Timestamp representation

BREAKING

KAFIL

PostgreSQL timestamptz values through Kafil's shared helper.

NAJM-THEME 0.1.1

ISO 8601 text in both package dialects.

Impact. Backfill cleanly, but update any direct database readers expecting Date values.

11. Preset primary key on PostgreSQL

ALIGNED

KAFIL

UUID primary key with defaultRandom().

NAJM-THEME 0.1.1

najm-theme/pg also uses UUID with defaultRandom(). SQLite alone uses text.

Impact. Existing Kafil preset IDs survive unchanged.

12. Preset slug uniqueness for one scope

ALIGNED

KAFIL

Slug is globally unique in a single installation.

NAJM-THEME 0.1.1

Slug is unique per (scope_id, slug). Kafil uses one constant platform scope.

Impact. The uniqueness guarantee is equivalent for Kafil.

Runtime behavior and server rendering**13. Revision protocol**

ALIGNED

KAFIL

Positive revisions, row lock, retry, and HTTP 409 on mismatch.

NAJM-THEME 0.1.1

Positive revisions, row lock plus compare-and-swap, and THEME_REVISION_CONFLICT.

Impact. Same concurrency semantics; the typed package error improves client recovery.

14. Reset semantics

ALIGNED

KAFIL

Store null, return factory values, increment revision.

NAJM-THEME 0.1.1

Store null, return factory values, increment revision.

Impact. A later factory change reaches reset installations in both systems.

15. Invalid stored appearance

ALIGNED

KAFIL

Warn, serve factory appearance, preserve stored revision.

NAJM-THEME 0.1.1

Emit a sanitized diagnostic, serve factory appearance, preserve stored revision.

Impact. Fallback and later optimistic conflict behavior match.

16. RSC bootstrap architecture

ALIGNED

KAFIL

One module-scope createReactServerUiBootstrap instance with lazy in-process fetch and independent resources.

NAJM-THEME 0.1.1

createReactThemeBootstrap configures that same Najm Kit primitive.

Impact. Replace Kafil parsers/resources with package configuration; keep the lazy fetcher and diagnostic sink.

17. Public branding wire format

BREAKING

KAFIL

Four required non-null *Path properties plus revision.

NAJM-THEME 0.1.1

{ slots: Record<string, string | null>, revision }.

Impact. Change the public endpoint type, RSC return type, layouts, branding editor, and tests. Factory configuration can keep the four standard values non-null, but the contract permits null and custom slots.

18. Standard slots and inheritance

ALIGNED

KAFIL

sidebar expanded/collapsed, auth logo, auth hero; collapsed and auth logo inherit from expanded defaults.

NAJM-THEME 0.1.1

The same four standard keys and the same inheritance graph.

Impact. No semantic slot rename is required.

Branding assets

19. Accepted image input

BREAKING

KAFIL

PNG, JPEG, WebP, and AVIF. New GIF uploads are explicitly rejected. Successful inputs are normalized to WebP.

NAJM-THEME 0.1.1

PNG, JPEG, and WebP only; the byte probe does not accept AVIF or GIF.

Impact. Audit legacy .gif/.avif references and re-encode them. Do not describe GIF as a current accepted upload.

20. Per-slot size ceilings

CONFIGURABLE

KAFIL

2 MB logos and 5 MB hero.

NAJM-THEME 0.1.1

Defaults are 512 KB and 2 MB; consumer limits may be raised up to the package's 8 MB ceiling.

Impact. Declare Kafil's current limits in brandingSlots so existing legal assets remain legal.

21. Stored asset reference

BREAKING

KAFIL

A full served route path is stored.

NAJM-THEME 0.1.1

The slot record stores fileName, MIME type, byte count, and upload time.

Impact. Strip the route prefix and derive trusted metadata from actual bytes during backfill.

22. Asset serve route

BREAKING

KAFIL

/api/branding/assets/serve/:fileName.

NAJM-THEME 0.1.1

/api/branding/assets/:fileName with basePath empty and Kafil's /api server base.

Impact. The only path-segment change is removal of /serve. Keep a compatibility redirect through the rollback window.

23. Candidate lifecycle

ALIGNED

KAFIL

Write candidate, commit database reference, then unlink replaced files; referenced candidates cannot be deleted.

NAJM-THEME 0.1.1

Same ordering, plus grace-period orphan reconciliation.

Impact. Preserve ordering and adopt reconciliation as a net reliability gain.

Silent contract differences

24. Appearance merge granularity

BREAKING SILENT

KAFIL

The validator updates a curated editable surface inside theme/components and preserves unrelated component fields.

NAJM-THEME 0.1.1

Any mentioned root group - including components - replaces that complete group.

Impact. A partial components payload can return 200, advance the revision, and remove siblings. The current Kafil UI submits a full design, but the public API semantics still change.

25. Preset name/slug collision

BREAKING SILENT

KAFIL

A duplicate normalized slug returns HTTP 409.

NAJM-THEME 0.1.1

The service derives -2, -3, and later suffixes until the slug is free.

Impact. An administrator can receive a second near-identical preset without an error. Freeze rejection versus suffixing after the School review.

Contract decisions before adoption

1. Appearance merge policy

Recommendation: keep najm-theme's whole-root-group replacement because it is the only unambiguous way to remove keys. Require every Kafil caller to submit a complete group, add a regression proving sibling removal is intentional, and document that PUT is a group replacement rather than a deep merge. Do not silently preserve Kafil's former sub-field semantics behind an app-only adapter.

2. Preset collision policy

Recommendation: wait for School before changing the package. If either consumer relies on duplicate rejection, add an explicit package option such as reject versus suffix and configure Kafil to reject. If suffixing remains universal, the UI must display the effective result clearly and contract tests must cover exact-name and normalized-slug collisions.

3. Unset branding

Keep the package wire type nullable and dynamic. Configure non-null factory values for Kafil's four standard slots, then update consumers to read slots[key] and handle null for custom or intentionally empty slots. Do not flatten the package back into four fixed fields.

4. Built-in preset deletion

Kafil's schema comment says built-ins cannot be deleted, while ThemePresetService.remove deletes them deliberately. The package defaults to disallowing deletion. Freeze the intended policy, then align the Kafil comment, backend, UI capability, and tests in one adoption change.

5. Actor attribution

The package deliberately avoids an auth dependency and stores actor IDs as text. Decide whether Kafil accepts durable historical IDs, adds an app-owned foreign key in its migration, or projects a separate immutable audit identifier. Record this before generating schema changes.

Rollback-safe migration

The previous one-step proposal was unsafe because it dropped the seven legacy settings columns during cutover. Use the staged sequence below.

1

Freeze and clean the source boundary

Complete the School review, resolve the five contract decisions, and commit or isolate Kafil's existing dirty worktree. Record exact source SHAs.

Gate: No implementation begins from an unreproducible snapshot.

2

Prepare dependencies

Upgrade najm-mcp from 2.0.2 to at least 2.1.0, add najm-storage at least 2.2.0, select its provider, and register storage before theme(). Keep najm-kit at or above 2.9.0.

Gate: Server boots with all stable peer tokens and the selected storage provider.

3

Add package tables only

Compose najm-theme/pg into Kafil's schema and create the three package tables. Leave platform_settings appearance/branding columns and theme_presets untouched.

Gate: Forward and rollback migrations both succeed without data loss.

4

Backfill deterministically

Use scope_id='platform'. Copy appearance and revisions. Convert four branding paths into slot records after inspecting each file. Copy preset UUIDs unchanged and preserve names, slugs, designs, built-in flags, actors, and timestamps.

Gate: Counts, revisions, UUIDs, slugs, and resolved output match before cutover.

5

Shadow-compare

Read legacy and package projections side by side in a verification command. Compare normalized appearance, resolved four-slot branding, preset rows, assets, and revisions without serving package output yet.

Gate: Every live row and referenced file matches or has an approved exception.

6

Cut over contracts

Register theme({ basePath: " }), switch the RSC bootstrap, update branding slots wire shape, move preset clients to /presets, update MCP names, and keep an asset-route redirect from /serve during the rollback window.

Gate: Production build, database tests, browser acceptance, and MCP discovery pass.

7

Burn in with rollback intact

Keep the legacy columns, table, and compatibility route for at least one release. Monitor diagnostics, 409 conflicts, asset misses, and preset behavior.

Gate: A rehearsed rollback can return to the previous application without reconstructing dropped data.

8

Remove legacy storage later

Only after acceptance, drop the seven platform_settings columns, retire the old theme_presets table, remove direct branding filesystem code, and end the /serve redirect.

Gate: Backup and isolated restore are verified before destructive cleanup.

Production data audit

Code inspection cannot prove what files and rows exist in production. Run these read-only checks before freezing the migration.

Branding references

Read the four *_path values from the platform row. Record nulls, duplicate references, extensions, and whether every path uses the managed branding prefix.

Actual asset bytes

For each referenced file: prove existence, byte size, decoded dimensions, detected MIME, and static-image status. Search specifically for legacy .gif and .avif references. Current uploads normally end as .webp.

Preset identity

Record count, UUID, slug, name, is_built_in, actor, timestamps, and design validity. Detect duplicate normalized names separately from duplicate slugs.

Appearance payloads

Validate the stored design against both Kafil and package parsers. Exercise a complete components group and a deliberately partial one in a disposable database.

External references

Search email templates, cached documents, automation, prompts, and bookmarks for /theme-presets, /branding/assets/serve, and legacy MCP names.

Acceptance record

- 1.** Focused contract tests for whole-group replacement, duplicate preset behavior, null slots, and typed revision conflicts.
- 2.** Migration and backfill tests against real PostgreSQL, including retry, rollback, duplicate slug, and missing-file cases.
- 3.** Package routes verified through Kafil's external /api base, including the temporary asset redirect.
- 4.** RSC production build proves one Najm design-provider identity and independent appearance/branding fallback.
- 5.** Browser acceptance covers login chrome, first-login chrome, settings page, sheet/dialog composition, save, reset, conflict recovery, presets, and asset replacement.
- 6.** MCP discovery proves the five published theme_* tools are present and old names are intentionally retired or mapped.
- 7.** Secret, scope, and dirty-worktree audits pass before commit, push, or deployment.
- 8.** Backup, isolated restore, deployment readiness, and application rollback are rehearsed before legacy-column removal.

Evidence and corrections

The corrected report resolves the factual and presentation defects in the prior nine-page export:

- The body now contains all 25 classified points and the displayed counts reconcile.
- The two silent behaviors are included as breaking contract points rather than referenced and omitted.
- Kafil's current upload path is described correctly: AVIF input is accepted, GIF input is rejected, and successful output is WebP.
- najm-theme's PostgreSQL preset ID is correctly identified as UUID, not text.
- The platform migration correctly counts seven legacy columns, not six.
- The external package asset route retains Kafil's /api server base; only the /serve segment changes.
- The required najm-mcp upgrade and new najm-storage dependency are explicit blockers.
- The migration is additive and rollback-safe; destructive cleanup is deferred.
- The Kafil source boundary is honestly described as HEAD plus a pre-existing dirty worktree.
- Section headings and final paragraphs are kept with content; there is no nearly-empty trailing page by design.

Primary source locations

Kafil packages/server/src/server.ts:19-26 - external /api base

Kafil packages/server/src/modules/settings/settingSchema.ts:37-47 - seven legacy columns

Kafil packages/server/src/modules/settings/appearanceValidator.ts:311-350 - curated merge

Kafil packages/server/src/modules/settings/themePresetService.ts:46-82, 85-109 - collision and deletion

Kafil packages/server/src/storage/managedImages.ts:130-165, 203-306 - MIME detection, GIF rejection, WebP output

Kafil apps/web/src/lib/serverLoader.ts:1-28 - module-scope RSC bootstrap

Kafil apps/web/src/lib/uiResources.ts:32-82 - current public payload parsers

Najm packages/najm-theme/src/contracts/appearance.ts:300-328 - whole-group replacement

Najm packages/najm-theme/src/contracts/presets.ts:98-120 - collision suffixing

Najm packages/najm-theme/src/contracts/branding.ts:75-115, 138-182, 287-290 - wire type, defaults, filename syntax

Najm packages/najm-theme/src/schema/pg.ts:49-113 - PostgreSQL table and UUID contracts

Najm packages/najm-theme/src/server/react.ts:94-143 - package RSC adapter

Najm packages/najm-theme/src/server/mcp/ThemeMcpTools.ts:91-165 - published MCP tool set

Najm packages/najm-theme/README.md:711-718 - required peer versions

Final recommendation

Treat this document as Kafil's half of the freeze, not approval to adopt. Review School against the same 25-point matrix, resolve the five contract decisions once for both consumers, then implement Kafil first through the staged migration.